

Fractions, Decimals, Percentages

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1 Fractions

1.1 Adding and Subtracting

If two fractions have the same denominator, add/subtract their numerator.

If they don't, multiply the top and the bottom by a number to make them have the same denominator.

Example 1.1 What is $\frac{3}{4} + \frac{7}{9}$

$$\frac{3}{4} + \frac{7}{9} = \frac{3 \times 9}{4 \times 9} + \frac{7 \times 4}{9 \times 4} = \frac{27}{36} + \frac{28}{36} = \frac{55}{36}$$

1.2 Multiplying

When multiplying two fractions, multiply their numerator together, and their denominator together.

Example 1.2 What is $\frac{3}{4} \times \frac{7}{9}$

$$\frac{3}{4} \times \frac{7}{9} = \frac{3 \times 7}{4 \times 9} = \frac{21}{36} = \frac{7}{12}$$

Look at the last step! We have divided both the top and the bottom number by 3. This is called **simplifying**. This is very important, because it makes your work a lot tidier.

1.3 Dividing

When we divide fractions, we will flip the one we are dividing upside down.

Example 1.3 What is $\frac{3}{4} \div \frac{7}{9}$

$$\frac{3}{4} \div \frac{7}{9} = \frac{3}{4} \times \frac{9}{7} = \frac{3 \times 9}{4 \times 7} = \frac{27}{28}$$

Problem 1.1 Why are we multiplying by flipping the fraction over?

Fraction Practice Problems

1. $\frac{1}{3} + \frac{2}{5}$
2. $\frac{7}{8} - \frac{1}{4}$
3. $2\frac{1}{2} + 1\frac{3}{4}$
4. $\frac{9}{5} - \frac{2}{3}$
5. $\frac{3}{4} \times \frac{2}{5}$
6. $1\frac{1}{2} \times \frac{4}{7}$
7. $\frac{5}{6} \times 2\frac{2}{3}$
8. $\frac{11}{4} \times \frac{8}{9}$
9. $\frac{2}{3} \div \frac{1}{4}$
10. $1\frac{1}{3} \div \frac{2}{5}$
11. $\frac{7}{9} \div 2\frac{1}{3}$
12. $\frac{5}{2} \div \frac{3}{8}$
13. $\frac{1}{2} + \frac{3}{7} - \frac{1}{14}$
14. $2\frac{1}{4} - 1\frac{1}{3} + \frac{1}{6}$
15. $\frac{5}{8} \times \frac{4}{15} \div \frac{1}{3}$
16. $1\frac{2}{3} \times \frac{9}{10} + \frac{1}{5}$
17. $\left(\frac{3}{4} + \frac{1}{8}\right) \times \frac{2}{7}$
18. $2\frac{1}{2} \div \frac{5}{6} - \frac{1}{4}$
19. $\frac{4}{5} + \frac{3}{10} - \frac{1}{2}$
20. $3\frac{1}{4} \times \frac{2}{13}$
21. $\frac{6}{7} \div \frac{3}{14}$
22. $1\frac{3}{5} + 2\frac{1}{4}$
23. $\left(\frac{2}{3} - \frac{1}{9}\right) \times \frac{3}{5}$
24. $\frac{10}{11} \div \frac{5}{22}$
25. $\frac{7}{12} + \frac{5}{18} + \frac{1}{4}$
26. $3\frac{1}{3} - 1\frac{5}{6}$
27. $\frac{3}{7} \times \frac{14}{9} \div \frac{2}{3}$
28. $2\frac{2}{5} \times 1\frac{1}{4} - \frac{1}{2}$

2 Decimals

Decimals are just place values, but going in the different direction.
For example, $0.27 = 0.2 + 0.07$ is just 2 tenths and 7 hundredths.

3 Percentages

Percentages are basically “How many bits out of 100 do I have”? Also, if we think about a 1 in decimal terms, 1 is equal to 100%.

Looking at our previous example then, $0.27 = \frac{27}{100} = 27\%$.

4 Conversions

4.1 Fraction to Decimal

1. $\frac{1}{2}$
2. $\frac{3}{4}$
3. $\frac{1}{5}$
4. $\frac{2}{5}$
5. $\frac{7}{8}$
6. $\frac{3}{10}$
7. $\frac{5}{3}$
8. $\frac{9}{4}$

4.2 Decimal to Percentage

1. 0.25
2. 0.5
3. 0.75
4. 0.1
5. 0.33
6. 0.875
7. 1.5
8. 2.25

4.3 Fraction to Percentage

1. $\frac{1}{4}$
2. $\frac{3}{5}$
3. $\frac{2}{3}$
4. $\frac{5}{8}$
5. $\frac{9}{10}$
6. $\frac{1}{6}$
7. $\frac{7}{4}$
8. $\frac{11}{5}$

4.4 Mixed Conversions

1. Convert 45% to a fraction
2. Convert 0.6 to a percentage
3. Convert $\frac{7}{10}$ to a decimal
4. Convert 82% to a decimal
5. Convert 0.125 to a fraction
6. Convert $\frac{5}{6}$ to a percentage
7. Convert 150% to a decimal
8. Convert 2.4 to a percentage
9. Convert $\frac{13}{8}$ to a decimal
10. Convert 225% to a fraction